

DAKSH KUMAR

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🌐 [LinkedIn](#)

🐙 [Github](#)

📁 [Portfolio](#)

Education

Indian Institute of Information Technology - Sri City

Bachelor of Technology in Computer Science and Engineering

CGPA: 8.98/10.0

Apr 2023 – June 2027

Sri City, Andhra Pradesh

Relevant Coursework

- Data Structures
- Cloud Computing
- Database Management
- Operating System
- Computer Networks
- Computer Architecture
- OOP's
- HPC
- Internet Security

Experience

SmallGroup

Aug 2025 – Jan 2026

Full-Stack Engineering Intern

Remote

- Engineered a full-stack stock scoring platform for **1,000+** U.S. equities using **Next.js** and **FastAPI**, enabling users to evaluate and compare stocks based on multiple financial and performance metrics.
- Designed and Implemented REST APIs to compute and serve stock scores, reducing client-side computation and improving page load times by **30%**.
- Constructed an automated environment management and **self-deployment system** across **5+** microservices on **GCP**, improving deployment consistency and rollback safety.
- **Automated CI/CD pipelines** and service orchestration, cutting manual deployment effort by **60%** and reducing configuration drift across **dev** and **prod** environments.
- Collaborated with a cross-functional team of engineers to standardize deployment practices, accelerating release cycles and improving production stability.

Enigma Club

Oct 2023 – May 2026

Co-Lead

College

- Coordinated and conducted Capture The Flag (CTF) sessions, helping participants develop skills in reverse engineering, cryptography, web exploitation, and binary exploitation.
- Developed a **RAM injection tool** for Linux, exploring low-level memory manipulation, process injection techniques, and system security vulnerabilities.

Projects

Event Driven Notification System (EDNS) | Go, Gin, Redis Streams, Pub/Sub, WebSocket, Prometheus, Docker

[Github](#)

- Designed and developed a **3-service microservice architecture** consisting of dedicated API, Worker, and WebSocket services, enabling asynchronous and real-time notification delivery.
- Implemented reliable message processing using **Redis Streams Consumer Groups** with automatic retries, Dead Letter Queue (DLQ), and **XAUTOCLAIM**-based recovery, supporting **at-least-once delivery** and seamless **horizontal worker scaling**.
- Integrated **2 notification channels** (SMTP Email and WebSockets) along with Prometheus metrics, health/readiness checks, structured logging, and Docker Compose for an observable production-ready deployment.

Modern Cloud-Native Kerberos | Python, Flask, WebSockets, React, Docker, PKINIT, Cryptography

[Github](#)

- Designed and Integrated a **passwordless Kerberos authentication system** using PKINIT and X.509 certificates, eliminating password-based attacks and reducing credential compromise risk by **100%**.
- Engineered a **high-availability Primary-Replica KDC** architecture with automated database replication and failover, supporting continuous authentication with **zero downtime**.
- Modernized Kerberos into a **cloud-native microservice architecture** using Docker, REST APIs, and a React UI, reducing manual provisioning effort by **60%** and enabling secure multi-device onboarding.

Pinak | TypeScript, Node.js, Express, PostgreSQL, Prisma, React, React Native, Turborepo

[Github](#)

- Engineered a **backend-first e-commerce platform** serving **2 client applications** (Web and Mobile), with a modular **Controller-Service-Repository** architecture that isolates HTTP handling, business logic, and persistence across domains.
- Built a **secure authentication and session system** supporting **2 authentication flows** (password and Google OAuth), using Argon2 hashing, access tokens and persistent refresh-token, token rotation, revocation, and **refresh-token reuse detection**.
- Designed **role-aware API boundaries** separating customer-facing and administrative operations, with centralized authorization, Zod-based request validation, standardized error handling, rate limiting, and shared TypeScript contracts to maintain consistency across backend and frontend clients.
- Containerized and deployed the **application on AWS EC2** using **Docker Compose**, configuring **Caddy as a reverse proxy** for HTTPS and routing, and integrating **Amazon S3** for scalable object storage and media delivery.

Technical Skills

Languages: C/C++, Go, Java, JavaScript, TypeScript, Python, HTML/CSS, SQL, BashScript

Technologies: Node.js, Gin, Next, React, Hono.js, React Native, Prisma, FastAPI, Turborepo, Pub/Sub

Databases: MongoDB, MySQL, PostgreSQL, SQLite, Redis

Cloud & DevOps: AWS, GCP, Docker, Kubernetes, Cloudflare Worker, NGINX, CI/CD, Prometheus, Grafana

Developer Tools: Git, Vim/Neovim, GDB, Wireshark, Postman, Linux System Calls